

Famous Hacker:

Michael Calce

Borne:

1884 in Montreal Quebec, Canada

Early Age:

After his parents separation his father bought him a computer at the age of Six (6). In his early hacking activities he has cheated the AOL (America Online an internet services provider) 30 days free trail and continually using internet services as an administrative.

Hacking Group:

He has joined the TNT hacking group or also known as TNT/Phorce.

Hacking Name:

“Mafiaboy”

Famous Activities:

He has hacked multiple famous websites in February 2000 at the age of just 15 years like Yahoo, Amazon, eBay, Dell, and CNN. This project was called Revolt and in this project he/his team used DDOS attack to shout down the world’s largest commercial websites which estimated the cost of \$1.2 Billion in global economy damage.

Failed Attacks:

He also attempted to launch attacks against nine of the thirteen root name servers.

How he was caught:

He was identified after boasting about his attacks in IRC chat rooms and claiming to have downed Dell's website before that specific attack had been publicly reported.

Arrest & Jail:

He was arrested by the Royal Canadian Mounted Police (RCMP) in April 2000 after being monitored by the FBI.

Due to being a young offender, he was sentenced on September 12, 2001, to **eight months in a youth detention center** (open custody), one year of probation, and a small fine (\$1,000).

After Release:

After releasing from the jail he has become a “White-hate” hacker, using his knowledge to help corporations identify vulnerabilities in their systems.

Also, he found his own company called **Optimal Secure** and authored a book called “Mafiaboy” how he Cracked the internet.

Domains of Cybersecurity

There are eight domains of Cybersecurity defined by CISSP (Certified System Security Professional) which are listed below:

1. Security and Risk Management
2. Asset Security
3. Security Architecture and Management
4. Communication and Network Security
5. Identity and Access Management
6. Security Assessment and testing
7. Security Operations
8. Software development Security

Let me explain each domain in detail.

- 1. Security and Risk Management:** It defines the security goals and objectives, risk mitigations, compliance, business continuity and the law.
 - a. Professionals in this domain act as the bridge between the technical teams and the executive leadership. Professionals' responsibilities are Policy Development, Risk Assessment, Compliance and Legal (Assessing security posture), Third-Party/Vendor Risk Management, Security Governance, Security Governance, Security Awareness Training and Incident Response Planning.
 - b. In required Technical skills, understanding of networks how networks are built to know where controls are needed. Knowledge of authentication protocols, difference between hashing, symmetric, and asymmetric encryption to choose the right control for data at rest and in transit, familiarity with shared responsibility models in AWS, Azure, or Google Cloud, and data loss prevention and threat modeling.
 - c. He/she must possess the communication skills, critical thinking, negotiation, ethics and integrity, and presentation skills.
 - d. Tools used in this domain are GRC Platforms (Governance, Risk, and Compliance), AuditBoard, Business Continuity Software, Policy Management Tools, Third-Party Risk Management (TPRM), and excel and power point.
 - e. A few certifications are used in this domain are, CompTIA Security+, Certified in Risk and Information Systems Control (CRISC), Certified Information Security Manager (CISM), Certified Information Systems Security Professional (CISSP), Certified Information Systems Auditor (CISA), and Certified in the Governance of Enterprise IT (CGEIT).

- f. A number of Career progressions are available like IT Auditor / Security Analyst (L1), Risk Analyst, IT Security Manager, Chief Information Security Officer (CISO), Chief Risk Officer (CRO) and Director of Compliance.
- 2. Asset Security:** It is the domain focused on identifying, classifying, and handling an organization's information assets (data and systems) throughout their lifecycle.
 - a. Its key responsibilities are Data Classification (Public, Internal, Confidential, Restricted), Inventory Management, Data Handling, Privacy, and Media Sanitization.
 - b. It required a few technical skills like Data Loss Prevention (block unauthorized data transfers), Encryption Management, Cloud Access Security Brokers, Mobile Device Management, and Digital Rights Management.
 - c. It include following soft skills Attention to Detail, Collaboration, Policy Enforcement, and Documentation.
 - d. For Asset security a few tools are used like ServiceNow, Snipe-IT, Microsoft Information Protection (MIP), Titus, Varonis, Microsoft Intune, and VMware Workspace One.
 - e. Certifications required like CISSP, Certified Data Privacy Solutions Engineer, and Certified Information Privacy Professional.
 - f. In career progress a number of rolls are available like Asset Management Technician, Data Security Analyst, Privacy Analyst, Chief Privacy Officer, and Chief Data Officer.
- 3. Security Architecture and Management:** It is the design, implementation, and management of security structures (networks, systems, software) to ensure an organization's defenses are built on a solid, resilient foundation.
 - a. Designing secure network and system architectures; selecting and implementing security tools (firewalls, SIEM); conducting cryptographic key management; and performing security control testing.
 - b. It required Deep knowledge of network protocols, firewalls, IDS/IPS, VPNs, encryption standards (PKI), identity management (IAM), and cloud security architecture (AWS/Azure).
 - c. Soft skills it required are creative problem-solving to bypass limitations, and the ability to document complex designs clearly for engineers
 - d. A number of tools are used like Cisco/Juniper firewalls, Wireshark (analysis), Terraform (Infrastructure as Code), AWS/Azure security tools, and enterprise password vaults (CyberArk, BeyondTrust).
 - e. A few certifications are required CISSP (Domain 3), Certified Information Systems Security Professional (CISSP) concentration, Certified Cloud Security Professional (CCSP), and vendor-specific certs like Cisco CCNP Security or AWS Security Specialty.
 - f. It has multiple career opportunities and growth like Network Security Engineer, Security Architect, Cloud Security Architect, Principal Engineer, Distinguished Engineer
- 4. Communication and Network Security:** The protection of data in transit across networks, focusing on designing secure network infrastructure, isolating threats, and ensuring secure connections (wired/wireless/cloud).
 - a. Key responsibilities include Configuring firewalls and routers, implementing VPNs and network access controls (NAC), monitoring for intrusions; and segmenting networks to contain breaches.

- b. It required Expertise in TCP/IP, OSI model, routing/switching (BGP/OSPF), firewall rules (ACLs), IDS/IPS, VPN protocols (IPsec/SSL), and network monitoring tools.
 - c. Soft skills required Analytical thinking for troubleshooting complex traffic flows; meticulous documentation of network changes; and calmness under pressure during network outages.
 - d. Required knowledge of tool of Cisco/Palo Alto/Check Point firewalls, Wireshark, SolarWinds, Nagios, Snort/Suricata (IDS), and Zscaler/Netskope (Secure Web Gateways).
 - e. Required certifications are CISSP (Domain 4), Cisco CCNP Security, CompTIA Network+/Security+, and Certified Network Defender (CND).
 - f. Its career path is Network Administrator, Network Security Engineer, Network Architect, Security Consultant. Growth into Cloud Networking (AWS VPC/Azure VNet) or DevSecOps.
- 5. Identity and Access Management:** It is the policy-driven discipline of ensuring the right people have right access to right resources at the right time.
- a. Professional's key responsibilities are Managing user identities (provisioning/deprovisioning), enforcing authentication (MFA), defining roles, and conducting access recertification audits.
 - b. This domain required technical skills are Knowledge of SSO protocols (SAML, OAuth, OpenID Connect), LDAP directories, scripting for automation, and privileged access management.
 - c. Soft skills required for this domain are Patience for user password issues, attention to detail for audit trails, and ability to design workflows for HR and IT coordination.
 - d. Tools used are Okta, Ping Identity, Microsoft Entra ID (Azure AD), CyberArk (PAM), SailPoint (Identity Governance), and RSA SecurID
 - e. Related certifications are CISSP (Domain 5), Certified Identity and Access Manager (CIAM), Okta Certified Professional, and vendor-specific Azure AD/AWS IAM certs.
 - f. Career growth includes IAM Analyst, IAM Engineer, IAM Architect, Identity Management Director.
- Growth into Identity Governance or Privileged Access Management (PAM) specialist.
- 6. Security Assessment and testing:** It defined that the continuous validation of security controls through hands-on testing, audits, and vulnerability assessments to verify that defenses work as intended.
- a. Key responsibilities includes planning and conducting penetration tests, running vulnerability scans, performing social engineering simulations, and reviewing security logs and code.
 - b. Its required technical skills of proficiency with scanning tools (Nessus), penetration testing frameworks (Metasploit, Burp Suite), scripting (Python, Bash), and log analysis (SIEM).
 - c. Meticulous documentation for reporting findings, analytical mindset for correlating vulnerabilities, and clear communication to explain risks to technical teams.
 - d. Tools used like Nessus/Tenable.io (vulnerability scanning), Metasploit (exploitation), Wireshark (packet analysis), Burp Suite (web app testing), and Cobalt Strike (adversary simulation).

- e. Certifications used like Offensive Security Certified Professional (OSCP), GIAC Penetration Tester (GPEN), and Certified Ethical Hacker (CEH).
 - f. This domain's career have multiple positions like Vulnerability Analyst, Penetration Tester, Red Team Lead, and Security Testing Manager.
Growth into specialized roles like Reverse Engineer, Exploit Developer, or Purple Team Lead.
- 7. Security Operations:** The day-to-day execution of security tasks to protect an organization, focusing on monitoring, detection, incident response, and investigation of security events.
- a. Key responsibilities include Monitoring SIEM alerts, triaging incidents, conducting forensic investigations, managing threat intelligence, and implementing prevention measures like EDR.
 - b. Key responsibilities include Monitoring SIEM alerts, triaging incidents, conducting forensic investigations, managing threat intelligence, and implementing prevention measures like EDR.
 - c. Required proficiency with SIEM tools, log analysis, EDR platforms, basic scripting (Python/PowerShell), and digital forensics techniques.
 - d. Soft skills required like Rapid critical thinking under pressure for incident response; clear and concise communication for escalation; and meticulous attention to detail for chain of custody.
 - e. In this roll tools used like Splunk/Elastic (SIEM), CrowdStrike/SentinelOne (EDR), TheHive (case management), Wireshark (analysis), and Autopsy/FTK (forensics).
 - f. Certification like GIAC Certified Incident Handler (GCIH), CompTIA Security+/CySA+, and Certified SOC Analyst (CSA).
 - g. Career growth includes SOC Analyst L1/L2/L3, Incident Responder, Threat Hunter, SOC Manager, and Director of Security Operations. Growth into DFIR (Digital Forensics and Incident Response) specialist or threat intelligence lead.
- 8. Software development Security:** It is the integration of security practices into the software development lifecycle (SDLC) to build secure applications by design, not as an afterthought.
- a. Key responsibilities are Conducting security code reviews, performing application security testing (SAST/DAST), establishing secure coding standards, and managing vulnerabilities in third-party libraries.
 - b. Required technical knowledge of programming languages (Python, Java, .NET), understanding of common web vulnerabilities (OWASP Top 10), and proficiency with application security testing tools.
 - c. Collaboration with developers (not confrontation), ability to provide constructive feedback, and translating security requirements into developer-friendly language.
 - d. Tools used like SonarQube (SAST), OWASP ZAP/Burp Suite (DAST), Snyk/Dependency-Check (SCA), and Git for version control integration.
 - e. Certifications used like Certified Secure Software Lifecycle Professional (CSSLP), GIAC Web Application Penetration Tester (GWAPT), and Offensive Security Web Expert (OSWE).
 - f. AppSec Analyst, Application Security Engineer, DevSecOps Engineer, Application Security Architect.

Growth into Product Security Lead or DevSecOps Director.

References:

- <https://moxso.com/blog/mafiaboy-the-boy-who-took-over-the-internet>
- https://en.wikipedia.org/wiki/Michael_Calce
- <https://www.secureworld.io/industry-news/video-inside-the-mind-of-hacker>
- <https://github.com/JasonTeixeira/CISSP-Exam-Prep>
- <https://niccs.cisa.gov/training/catalog/cybrary/cissp-domain-3-security-architecture-and-engineering>
- <https://www.oreilly.com/library/view/cissp-cert-guide/9780135343869/ch08.xhtml>
- <https://www.sciencedirect.com/science/article/pii/B9780128112489000073?via%3Dihub>
- <https://netwrix.com/en/resources/blog/cissp-exam-changes-2024/>
- chat.deepseek.com/chat/s/